CALPHY+

Electric Forces and Electric Field

ECE, Instructor

OUTLINE

- Electric Charge
- Coulomb's Law
- Electric Field Intensity

Electric Charge

- In physics, charge, also known as electric charge, electrical charge, or electrostatic charge is a characteristic of a unit of matter that expresses the extent to which it has more or fewer electrons than protons.
  - A proton has a positive charge.
  - An electron has a negative charge.
  - A neutron has no net electric charge.

INTRODUCTION

- An **atom** consists of a small, relatively massive nucleus that contains particles called **protons** and **neutrons**.
- A proton's mass is 1.673×10^{-27}kg, and neutron has slightly greater mass of 1.675×10^{-27}kg.
- Surrounding the nucleus is a diffuse cloud of orbiting particles called **electrons**(9.11×10^{-31}kg).

INTRODUCTION

- A **proton** has a **positive charge**, and an **electron** has a **negative charge**. A neutron has no net electric charge.
- Experiment reveals that the magnitude of the charge on the proton *exactly equals* the magnitude of the charge on the electron.

Electric Charge

- In physics, charge, also known as electric charge, electrical charge, or electrostatic charge is a characteristic of a unit of matter that expresses the extent to which it has more or fewer electrons than protons.
- The most basic quantity in an electric circuit
- An electrical property of the atomic particles of which matter consists
- SI unit: **Coulombs (C)**

Electric Charge

- charge *e* of an electron = -1.602×10^{-19} C
- charge of a proton = 1.602×10^{-19} C
- Neutron has no charge.

Sample Problem

- How many electrons are there in one coulomb of negative charge?

Electric Charge

1. The coulomb is a large unit for charges.

$1\ C = 1/(1.602 \times 10^{-19}) = 6.24 \times 10^{18}$ electrons

Thus realistic or laboratory values of charges are on the order of pC, nC, or μC.

Electric Charge

2. According to experimental observations, the only charges that occur in nature are integral multiples of the electronic charge.

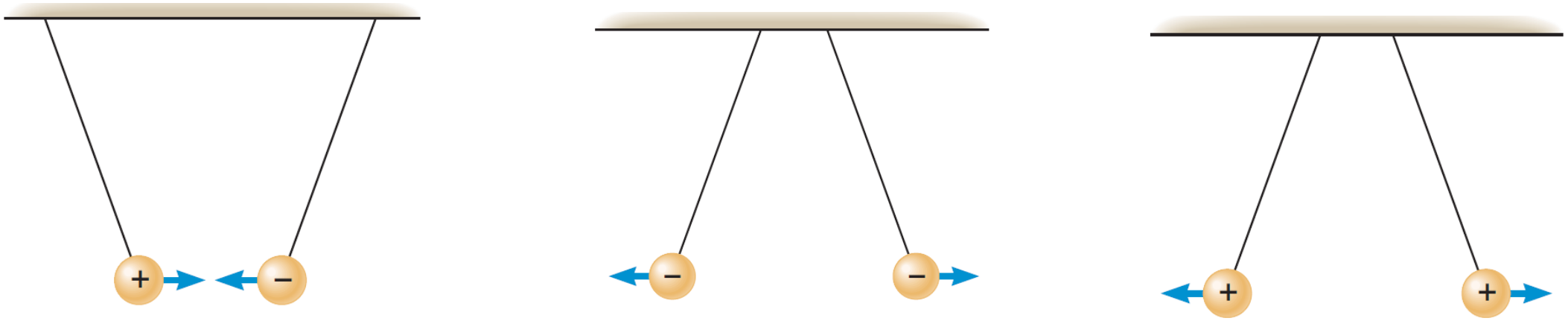
$e = -1.602 \times 10^{-19}\,\mathrm{C}$

Electric Charge

3. The ***law of conservation of charge*** states that charge can neither be created nor destroyed, only transferred. Thus the algebraic sum of the electric charges in a system does not change.

Electric Charge

- Two electrically charged objects exert a force on one another:



- **Like charges repel and unlike charges attract each other.**

Electric Charge

- Electric charge can not only exist on an object, but it can also *move through* an object.
- Substances that readily conduct electric charge are called **electrical conductors**.
  - Metals such as copper, aluminum, silver, and gold are excellent electrical conductors and, therefore, are used in electrical wiring.

Electric Charge

- Materials that conduct electric charge poorly are known as **electrical insulators**.
  - Common electrical insulators are rubber, many plastics, and wood. Insulators, such as the rubber or plastic that coats electrical wiring, prevent electric charge from going where it is not wanted.

COULOMB'S LAW

- The magnitude F of the electrostatic force exerted by one point charge q_1 on another point charge q_2 is directly proportional to the magnitudes q_1 and q_2 of the charges and inversely proportional to the square of the distance r between them:

$$\boldsymbol{F = k\frac{|q_1||q_2|}{r^2}}$$

- where k is a proportionality constant:
- $k = 8.99 \text{ x } 10^9 \text{ N m}^2/\text{C}^2$ in SI units.

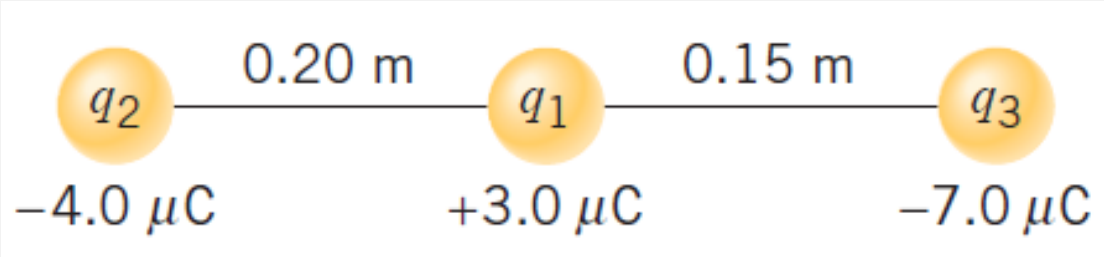
Sample Problem

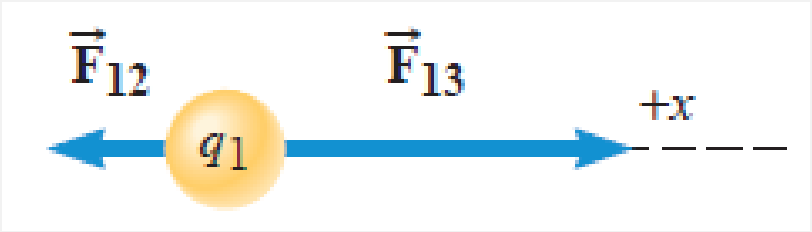
1. Two objects, whose charges are +1.0 and -1.0 C, are separated by 1.0 km. Compared to 1.0 km, the sizes of the objects are small. Find the magnitude of the attractive force that either charge exerts on the other.

COULOMB'S LAW

- **The force on a point charge due to two or more other point charges.**

  Example1: Determine the magnitude and direction of the net electrostatic force on q_1.

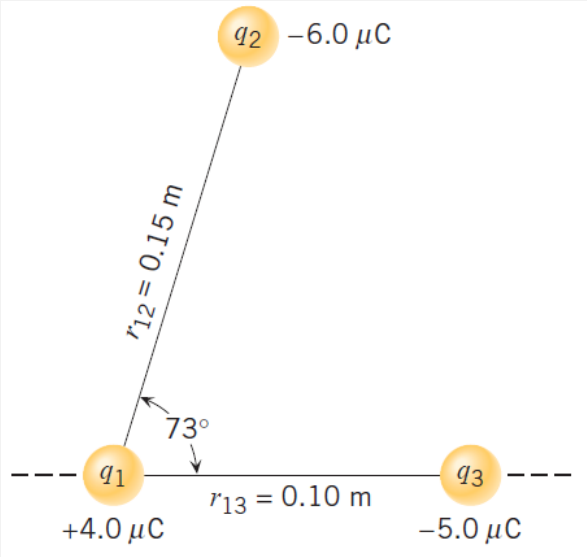




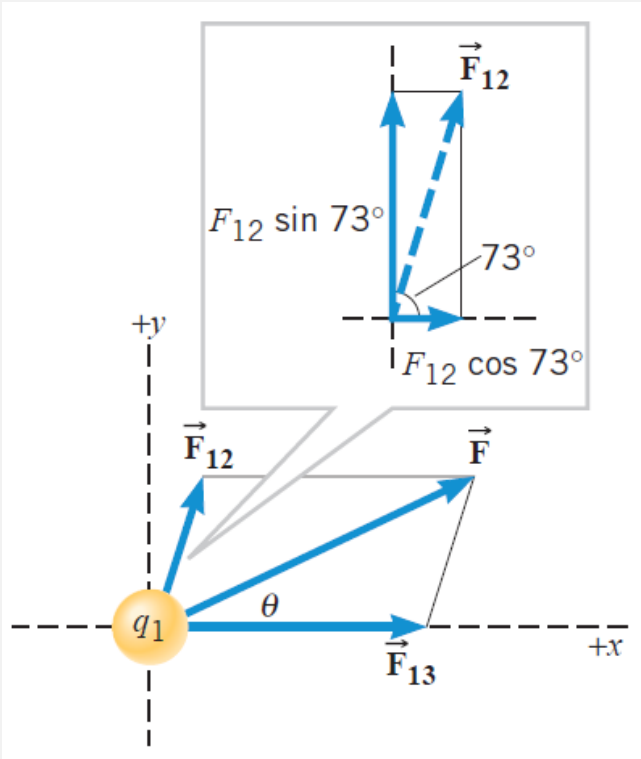
COULOMB'S LAW

- **The force on a point charge due to two or more other point charges.**

Example2: find the magnitude and direction of q_1



F=23
Deg = 24°

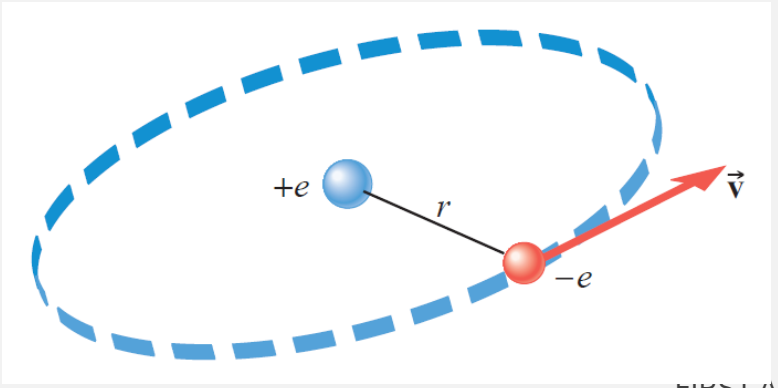


Centrifugal Force of a Particles

In the Bohr model of the hydrogen atom, the electron (charge $-e$) is in a circular orbit about the nuclear proton (charge $+e$) at a radius of 5.29×10^{-11} m. The mass of the electron is 9.11×10^{-31} kg. Determine the speed of the electron.

Note! Formula for centrifugal force

$$F_c = mv^2/r$$



Given: $m_e = 9.11 \times 10^{-31}$ kg

$r = 5.29 \times 10^{-11}$ m

Find: V

Soln: $F_c = mV^2/r$

$$V = \sqrt{\frac{F_c r}{m}}$$ ✓ ✓

$V = 2.188 \times 10^6$ m/s

$$F_c = F = k\frac{|q_1||q_2|}{r^2}$$

$$F = \left(8.99 \times 10^9 \frac{N \cdot m^2}{C^2}\right)$$

$q_1 = -1.602 \times 10^{-19}$ C

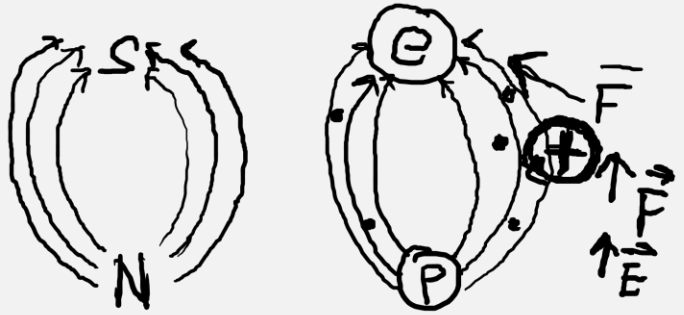
$q_2 = 1.602 \times 10^{-19}$ C

$F = 8.245 \times 10^{-8}$ N

Electric Field

- **Electric Field** $\vec{E}$ that exists at a point is the electrostatic force $\vec{F}$ experienced by a small test charge q_0 placed at that point divided by the charge itself:

$$\vec{E} = \frac{\vec{F}}{|q_o|}$$



- The electric field is a vector, and its direction is the same as the direction of the force $\vec{F}$ on a positive test charge.
- ***SI Unit of Electric Field:*** newton per coulomb (N/C)

Electric Field Intensity

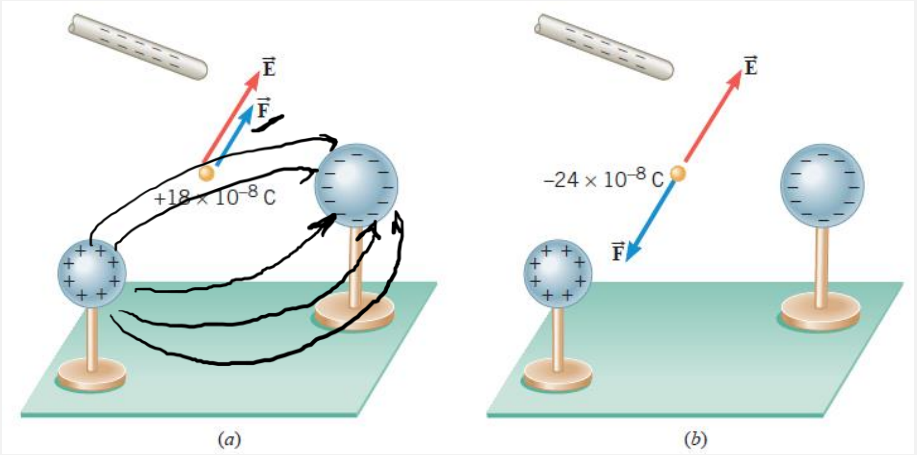
- The charges on the two metal spheres and the ebonite rod create an electric field $\vec{E}$ at the spot indicated. This field has a magnitude of 2.0 N/C and is directed as in the drawing. Determine the force on a charge placed at that spot, if the charge has a value of **(a)** q_0=+18x10^{-8} C and **(b)** q_0=-24x10^{-8} C.

$$\vec{E} = \frac{\vec{F}}{|q_0|} \left(\frac{N}{C}\right)$$

a) $\vec{F} = |q_0|\vec{E}$

$\vec{F} = |18 \times 10^{-8}|\left(2 \frac{N}{C}\right)$

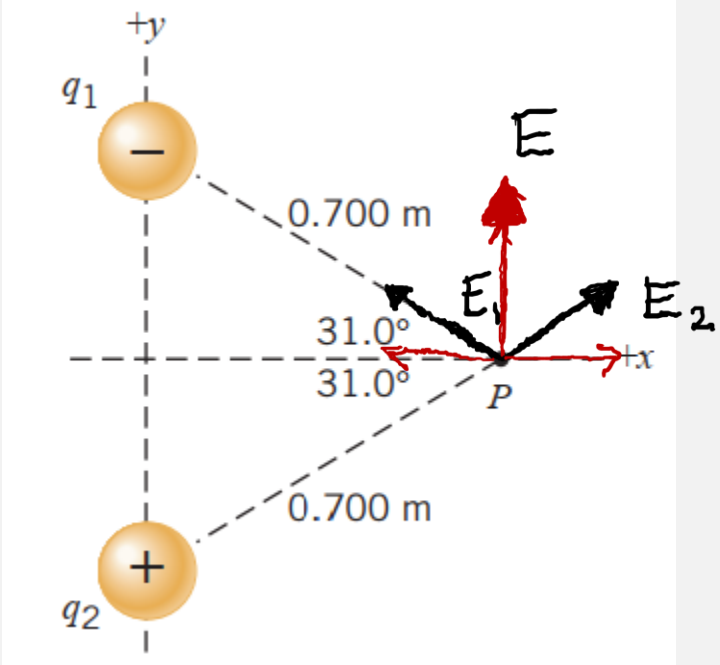
$\vec{F} = 3.6 \times 10^{-7}$ N



b) $\vec{F} = |-24 \times 10^{-8}|\left(2 \frac{N}{C}\right)$

$= 4.7 \times 10^{-7}$ N

- Two point charges are lying on the y axis as shown in figure. They are equidistant from the point P, which lies on the x axis. What is the electric field at P? $q_1 = -4.00\ \mu\text{C}$ and $q_2 = +4.00\ \mu\text{C}$.



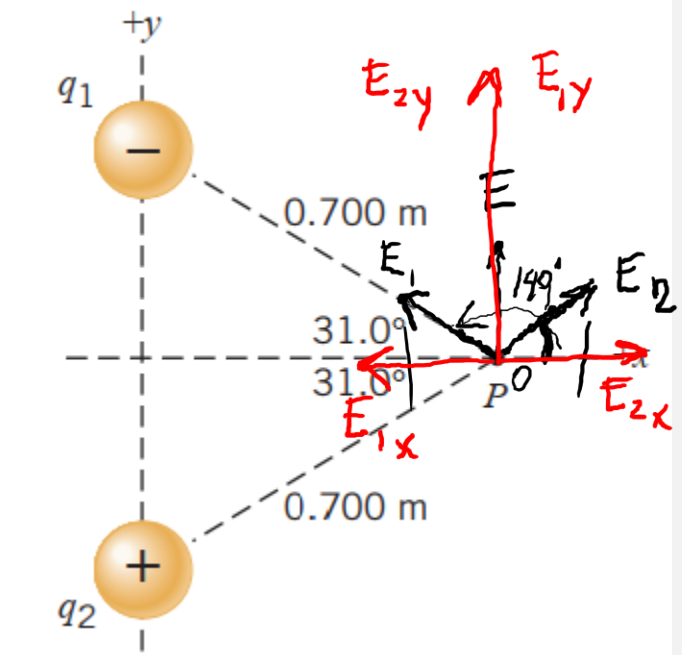
$$E = \frac{F}{q_0} \leftarrow \qquad E_1 = \frac{k\,q_1}{r^2} \qquad E_2 = \frac{k\,q_2}{r^2}$$

$$F = k\,\frac{|q_a||q_b|}{r^2}$$

$$= \frac{\left(8.99\times10^{9}\ \frac{N m^2}{C^2}\right)(4\,\mu C)}{(0.7\ m)^2}$$

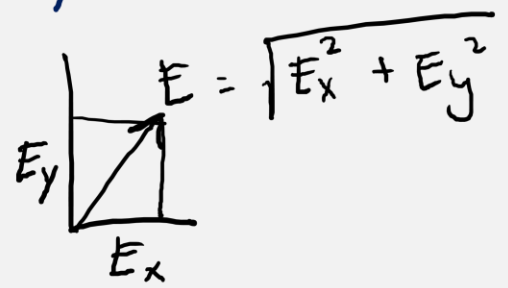
$$E_1 = 73.388\times10^{3}\ \frac{N}{C}$$

$$E_2 = 73.388\times10^{3}\ N/C$$



$$E = \sqrt{E_x^2 + E_y^2} = \sqrt{0^2 + E_y^2} = E_y$$

$$E = 75.595 \times 10^3 \ \frac{N}{C} \text{ along y-axis}$$



	x component	y component
E_1	$-E_1 \cos(180-31^\circ)$	$+E_1 \sin(180^\circ - 31^\circ)$
E_2	$+E_2 \cos(31^\circ)$	$+E_2 \sin(31^\circ)$
	0	75.595×10^3 N/C

THANK YOU!